Graphic organiser toolkit – Supporting critical and creative thinking

What are graphic organisers?

Graphic organisers are useful tools for assisting students to visually organise complex ideas and information. They are valuable in enhancing student learning and thinking by providing students with the tools to organise and represent their knowledge and understanding about concepts and to identify relationships between ideas.

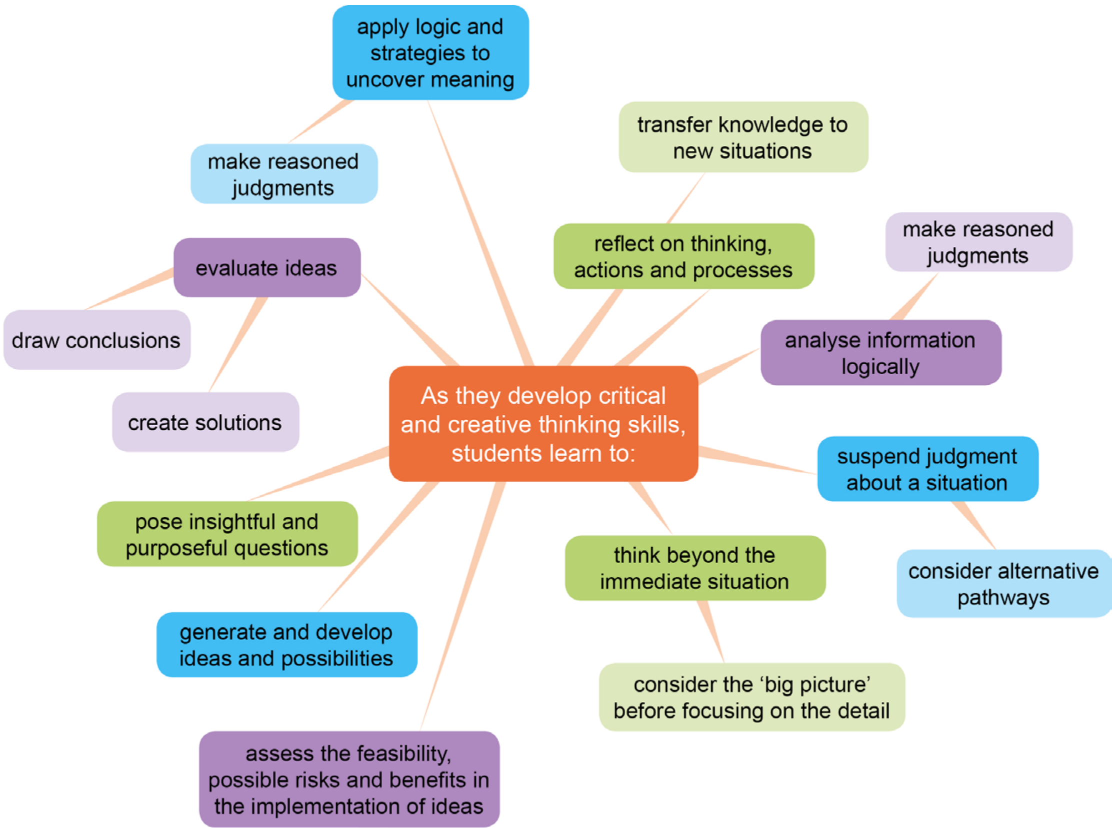
Graphic organisers allow students to represent the patterns and connections between information, ideas and concepts.

Supporting critical and creative thinking

Critical and creative thinking has been identified as one of the seven general capabilities in the Australian Curriculum. Critical and creative thinking refers to higher-order thinking. For example critical and creative thinking describes the top three levels of Bloom's Revised Taxonomy – analysing, evaluating and creating (Anderson, L and Krathwohl, D (eds.) 2001, *A Taxonomy for Learning, Teaching and Assessing: A Revision of Blooms Taxonomy of Educational Objectives*, Allyn & Bacon, Boston, MA.).

Students develop critical and creative thinking as they learn to generate and evaluate knowledge, ideas and possibilities, and use these when looking for new pathways or solutions. In learning to think broadly and deeply students learn to apply logic and imagination to direct their thinking for different purposes. In the context of the classroom, critical and creative thinking are essential to activities that require reason, imagination, originality and innovation.

Source: http://www.australiancurriculum.edu.au/GeneralCapabilities/Critical-and-creative-thinking



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Graphic organisers can support students to think critically and creatively and to record their thinking for a range of purposes. Graphic organisers can be an end in themselves or used as a stimulus for developing spoken and written arguments, reports, procedures and explanations about a variety of concepts and topics.

Graphic organisers and thinking

Graphic organisers make thinking visible. They allow students to clearly represent what they know and understand as well as identify the relationships between information and concepts.

Graphic organisers should be used as an integral component of thinking activities. These graphic tools support students to organise their thinking so it becomes an explicit part of the learning process.

Graphic organisers promote creative and critical thinking by providing students with the tools to visually:

- relate new learning to prior knowledge
- identify relationships between facts and concepts
- represent and evaluate complex ideas
- organise and analyse information
- generate new knowledge and understandings
- solve difficult problems
- make decisions
- reflect on their learning.

Selecting graphic organisers

The type of task will determine the type of graphic organiser that will be most appropriate. It is important to select graphic organisers that suit the learning and teaching purpose of the task. Students need to understand the purpose for using the graphic organiser, how the organiser is used, and the type(s) of thinking required to complete the task. Some examples include:

Teaching and learning focus	Graphic organiser
Brainstorm	Concept map, Y chart
Compare and contrast	Venn diagram, T chart
Link prior learning and new learning	TWLH chart
Sequence ideas, steps or process	Flow chart
Explore relationships	Cause and effect wheel
Analyse parts	PMI chart, SWOT analysis
Make decisions	SWOT analysis
Reflect on learning	Mind map, Plus/Minus/Interesting tool

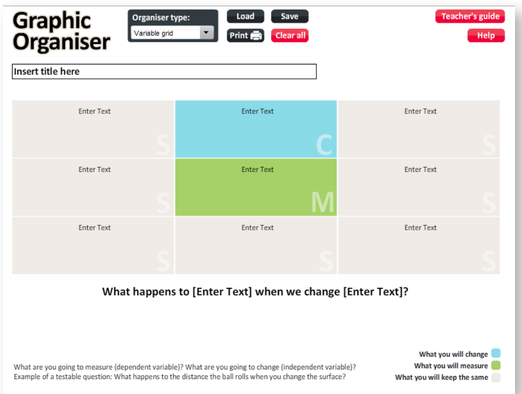
The graphic organiser needs to be explained and modelled across a number of different contexts before students use the graphic organiser for the first time. Then students should be provided with multiple opportunities to practise using the graphic organiser in different contexts and for a range of purposes. Eventually, the student should be able to independently select a specific graphic organiser to suit a type of task or style of thinking, or even design their own graphic organiser to suit a task.

Types of graphic organisers

The type of thinking or the style of activity should be used to determine the type of graphic organiser that is most suitable. The graphic organisers below are all available as part of the *Graphic organiser toolkit* and can be used to assist students to work and think in different ways.

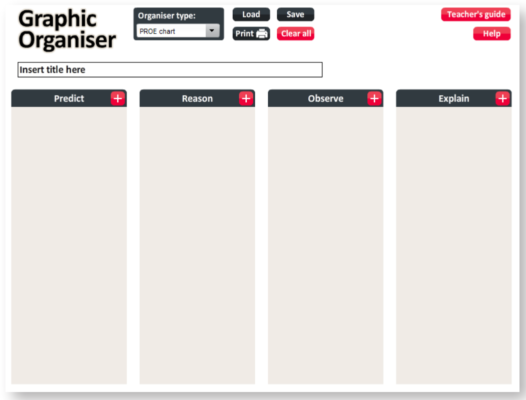
Variable grid

A Variable grid is used to help students **brainstorm ideas and generate questions** for fair testing in scientific investigations. For example: *What happens to the distance the ball rolls when we change the surface it rolls on?*



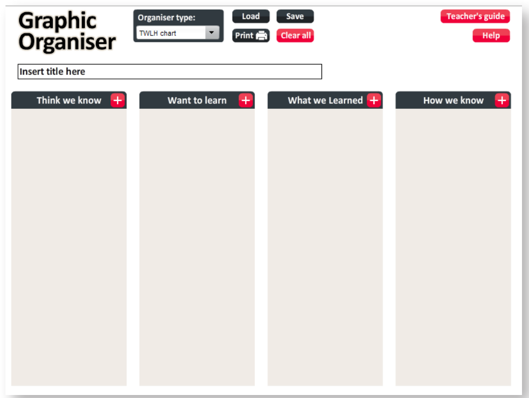
PROE chart

A PROE chart is used to **record** predictions, reasons to support predictions, observations for each prediction and explanations for why the results did or did not match the predictions (based on a comparison of predictions and observations). For example: *What happens when we drop a 1kg bag of feathers and a one kilogram bag of rocks from a height of 3 metres?*



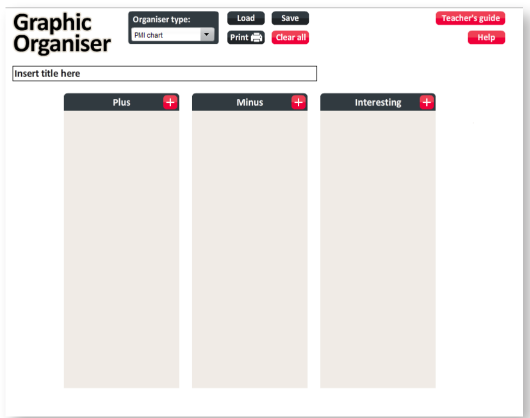
TWLH chart

A TWLH dhart is used to **organise and analyse** students' prior knowledge, determine the questions that students want to know the answer to, document what has been learned, and allow students to include evidence from investigations and learnings to explain and justify claims. It is designed to be used before, during and after the study of a new unit or concept. For example: *What do we know about electrical circuits? What do we want to know about electrical circuits? What have we learned about electrical circuits this term? Explain how you know this.*



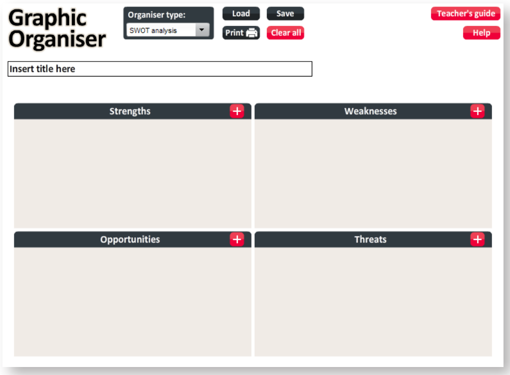
PMI chart

A PMI chart is used to **organise and analyse** the benefits, deficits and interesting points of an idea, topic, subject or proposal. It can be used for initial brainstorming, or at the conclusion of an investigation or task. For example: *Analyse the positive, negative and interesting points associated with using preserved DNA to recreate extinct animals such as dinosaurs.*



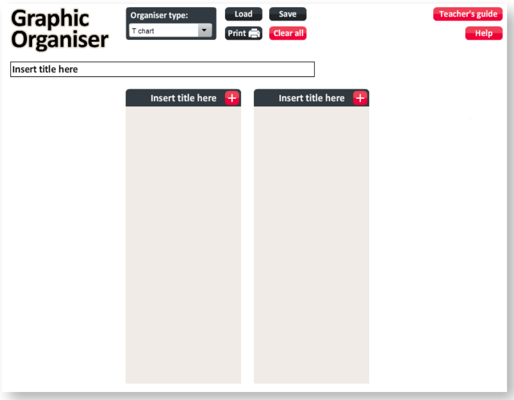
SWOT analysis

A SWOT analysis is used to **analyse and evaluate** the strengths, weaknesses, opportunities and threats of an idea. It is useful when used to assist students to **make decisions** about implementing actions or new ideas. For example: *Use the SWOT analysis tool to analyse and evaluate installing solar panels on the tuckshop. Use the SWOT analysis tool to analyse and evaluate the council using recycled water*.



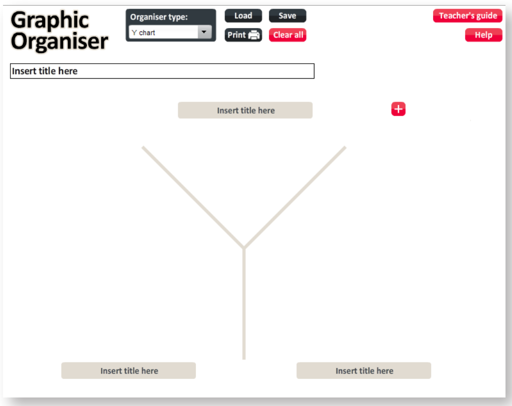
T chart

A T chart is used to **organise and/or compare** information or thinking about two facets of one topic (e.g. facts and opinions, pros and cons) or about two different concepts or subjects. For example: *Use a T chart to evaluate the pros and cons associated with using nuclear energy. Use a T chart to compare frogs and toads*.



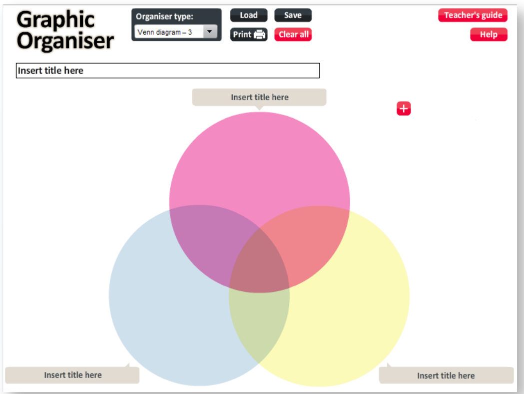
Y chart

A Y chart is used to **organise and analyse** three different qualities of a topic (e.g. looks like, sounds like, feels like). It can be used for individual or whole class brainstorming about an idea or concept. For example: *Describe what it would be like during an earthquake or tsunami. Describe a character in a book, including what the character looks like, sounds like, and how the character feels*.



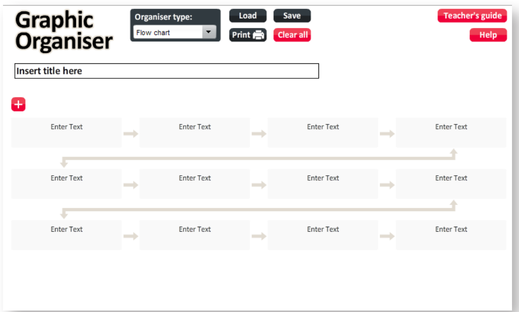
Venn diagram (2 circle)
Venn diagram (3 circle)

A Venn diagram is used to **compare and contrast** the similarities and differences between two or three items or topics. A Venn diagram organises information and ideas to illustrate differences between the topics in the parts of the circles that do not overlap; and similarities in the parts of the circles that do overlap. For example: *Examine the similarities and differences between nuclear and wind energy. Use a Venn diagram to compare a shark, a whale and a dolphin*.



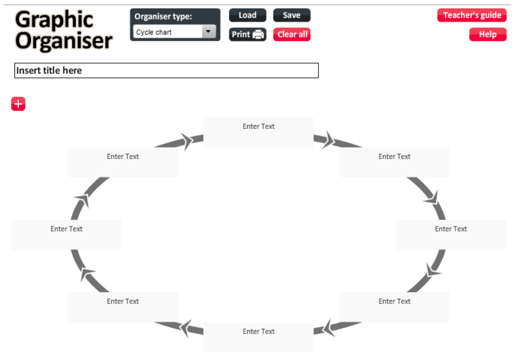
Flow chart

A flow chart is used to **organise and sequence** ideas, information or processes. It allows for each step in a process to be labelled and described. For example: *Use a flow chart to explain how an email gets from the sender to the recipient. Use a flow clart to show the process for how water gets from the dam to your tap. Show the glass recycling process using a flow chart. Use a flow chart to show how clouds are formed.*



Cycle chart

A cycle chart is used to show how items, events or stages are related to one another in an ongoing or repeating cycle. In making a cycle chart, the student must identify the main events in the cycle, how they interact with each other or what changes occur between events, and how the cycle repeats. For example: *Use a cycle chart to represent the life cycle of the frog. Show the water cycle using a cycle chart. Describe the life cycle of a flowering plant using a cycle chart*.



Related resources

Brown, David, *Using the CSF in the Middle Years*, Victorian Essential Learning Standards, http://vels.vcaa.vic.edu.au/support/usingcogorg.html
A resource explaining how and why to use various graphic organisers (referred to as cognitive organisers) in the middle years.

Ellis, Edwin, (2004). *Q&A: What's the big deal about graphic organizers?* http://mcs.monet.k12.ca.us/Academics/el712/Graphic%20Organizers/Q%20and%20A%20about%20G%20O.pdf
A professional reading providing information about graphic organisers including when and why to use them.

Graphic Organisers, Victorian Essential Learning Standards, http://vels.vcaa.vic.edu.au/support/domainsupport/thinking/organisers.html
This site provides information to help teachers select an appropriate graphic organiser to develop particular thinking skills or concepts. Links can be selected to locate related examples and templates.